Final Planning Document

Senior Design I

Sisyphus’ Schedule - Web App

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1. An overview/executive summary of your project (a paragraph or two).

    Our app, Sisyphus’ Schedule, is a fitness web app that aims to streamline workout planning and tracking through personalization and machine learning-generated workouts. Users create an account and profile with their gender, age, weight, experience level, and fitness goal (bulk, cut, gain strength, or muscle). The app generates or allows users to build weekly workout splits, such as push/pull/legs or upper/lower. It tailors exercises to the user’s preferences, enabling them to like or dislike movements and automatically swap disliked ones for suitable alternatives. Users can log sets, reps, and weights, track progress, and view upcoming workouts through an integrated calendar. Our app focuses on delivering a seamless workout generation and logging experience that adapts to user feedback and data by April 2026.

2. A concise statement of the MVP. this is one to 2 sentences.

    Our app will allow users to participate in generated workouts, create their own, as well as track the weights, and completion of these workouts. Additionally, our web app allows users to track calories, including carbs, proteins, and fats.

3. The current team (you!) with short 2 sentence bios as well as what "role" / lead you are each taking in the project
    - Tesla Norton - Project Originator & Frontend Designer: I originated the concept for Sisyphus’ Schedule and will manage the final presentation. I work with Lily regarding the user flow but focus on designing and styling the web app’s user interface to deliver the best user experience.
    - Lily Brongo - Web Designer, working with mockups and user flows to develop a functioning UI as well as implementing user testing by including questionnaires and feedback points.
    - Collin Davis - Machine Learning Model Engineer: I am creating our Machine Learning Model to generate workouts for the user based on a questionnaire they fill out when logging into the app. It will generate new workouts every month for the user and keep them engaged.
    - Thomas Hinckley - Backend developer: Integrating database and functionality for web application along with other backend tasks such as APIs and logic.

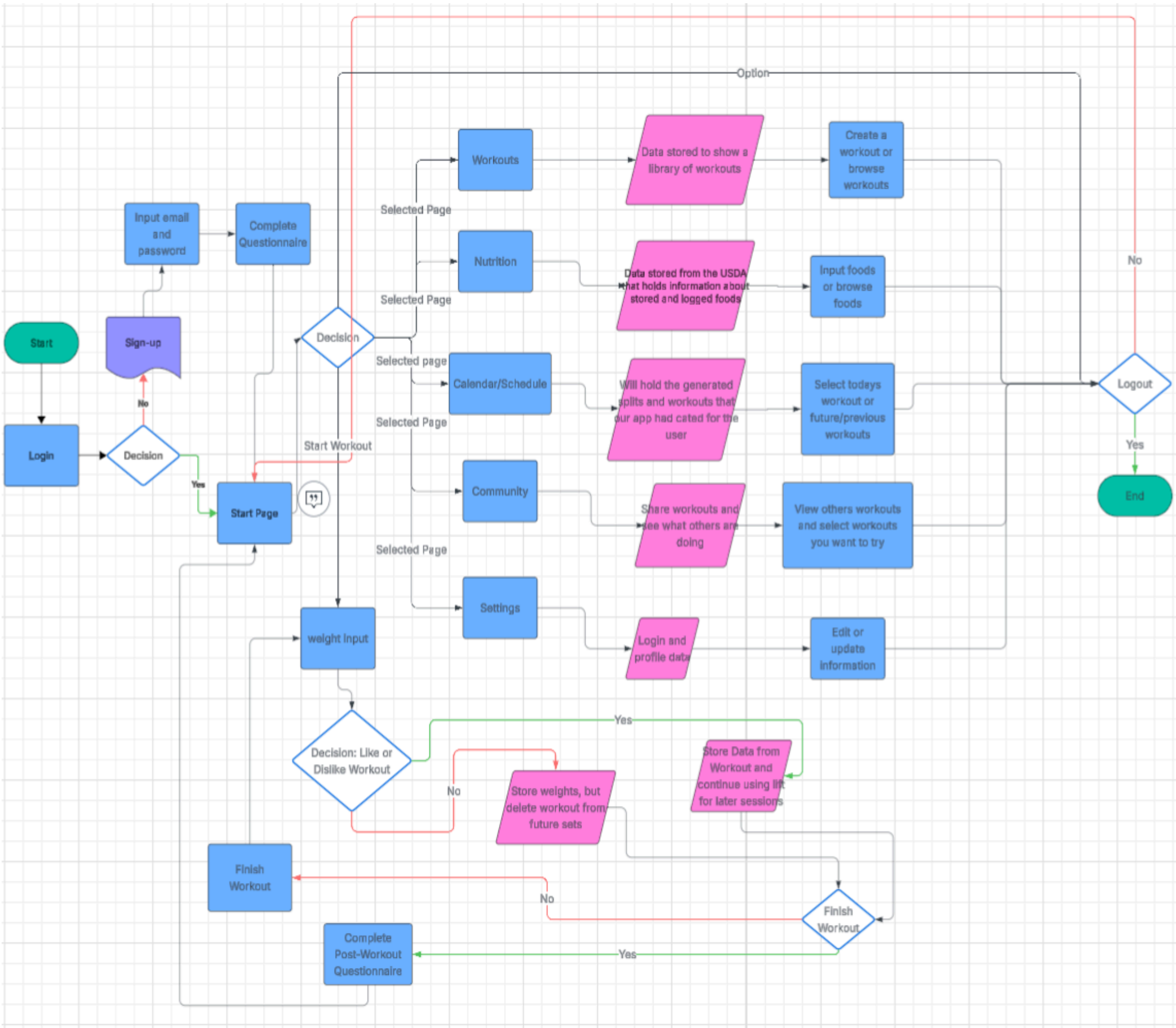
- Nolan Nachbar - Backend developer: Work with Thomas to connect the frontend to the database and integrate all the APIs (authentication, exercise data, and nutrition APIs) that we'll need.

4. A description and/or list of the functional requirements of your end product and/or minimum viable product (what must the final or MVP be able to do?). I will also help with deployment and backend architecture.
    - MVP:
        i. Users can create an account and sign in.
        ii. User Profile Setup: Collect gender, age, weight, experience level, and goal (bulk, cut).
        iii. Workout Split Generator: ML model can generate, or the user can create a weekly split (push/pull/legs, upper/lower, etc.).
        iv. Rating system: Allow users to like/dislike exercises.
        v. Exercise Customization: Allow users to swap disliked exercises and prompt alternatives based on the same muscle group.
        vi. Workout Logging: Let users log sets, reps, and weights for each exercise.

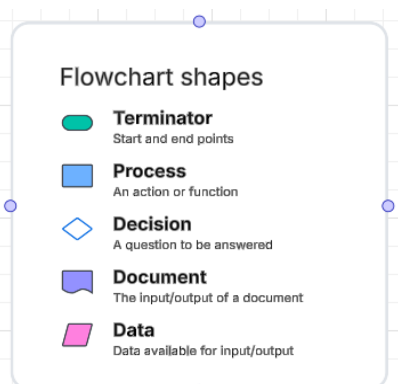
5. A High-Level Systems Overview [e.g. what are the major components in your system]; Diagrams and Text descriptions would be appropriate
    1. Workout generator
        a. Based on experience we will generate workouts that a user can like and dislike. This information will be stored in our database and then used for future workouts for the user.
        b. Within the workouts we will allow the user to create their own workout and save their weights for future references. The created workouts will allow for experienced users to have the customizability we want to provide.
        c. Provide intensity levels, recovery options, and post workout questionnaires to see how the workout generator was for the user.
        d. We will store weights, experiences, likes and dislikes, as well as projected workouts within our database (all will be accessible to the user).
    2. Nutrition Tracker
        a. Users will be able to log food and see the carbs, fats, and protein within their meals.
        b. Will use the FDA database for food that contains information regarding calories, proteins, fats, and carbs. This implementation will allow for better feedback to users and give the application a complete experience.
        c. Look to input calories surplus or deficit markers depending on implemented goals from users (weight loss, weight gain, maintenance).

3. MachineLearning - Subsystems:
    i. Authentication and User Profiles
        1. Dependencies:
            a. Database to store user data
            b. User model to save data such as workouts to user account
        2. Constraints:
            a. Security must be implemented
    ii. Workout Generator (ML-model)
        1. Dependencies:
            a. Database of exercises (Guide given to Josh Little)
            b. Build an algorithm based on the splits provided by Josh
            c. Data to train and test the model
            d. A server that can run the models
        2. Constraints:
            a. The quality of the model depends on the amount of training and test data we have.
            b. Finding a server that will have the computational power to run the model.
            c. Limited workout algorithm generator because of infinite possibilities
    iii. Workout Logging
        1. Dependencies:
            a. Database to store workout information
        2. Constraints:
            a. Finding a server big enough to store it all.
    iv. Nutrition and Macro
        1. Dependencies:
            a. Database to store user nutrition history
            b. Barcode scanner API
            c. Nutrition database API (USDA)
            d. Macro calculator
        2. Constraints:
            a. Must handle incomplete or approximate food entries
            b. Reliant on third-party nutrition APIs so we have to deal with uptime and rate limits.
    v. Calendar and Scheduling
        1. Dependencies:
            a. Integration with the workout logging subsystem
        2. Constraints:
            a. It could require more dependencies depending on whether we include time zones or other features.
    vi. Community / Social Features
        1. Dependencies:
            a. Database for friends list - Handles friends list, leaderboards, and shared workout templates.
        2. Constraints:
            a. Stretch goal, so time may not allow us to include this feature in time.

vii. Backend Infrastructure
    1. Dependencies:
        a. Frontend UI mostly complete to integrate functionality logic
        b. Database - Supabase/PostgreSQL Database
viii. Frontend UI Design
    1. Dependencies:
        a. Framework - CSS and Javascript on top of HTML

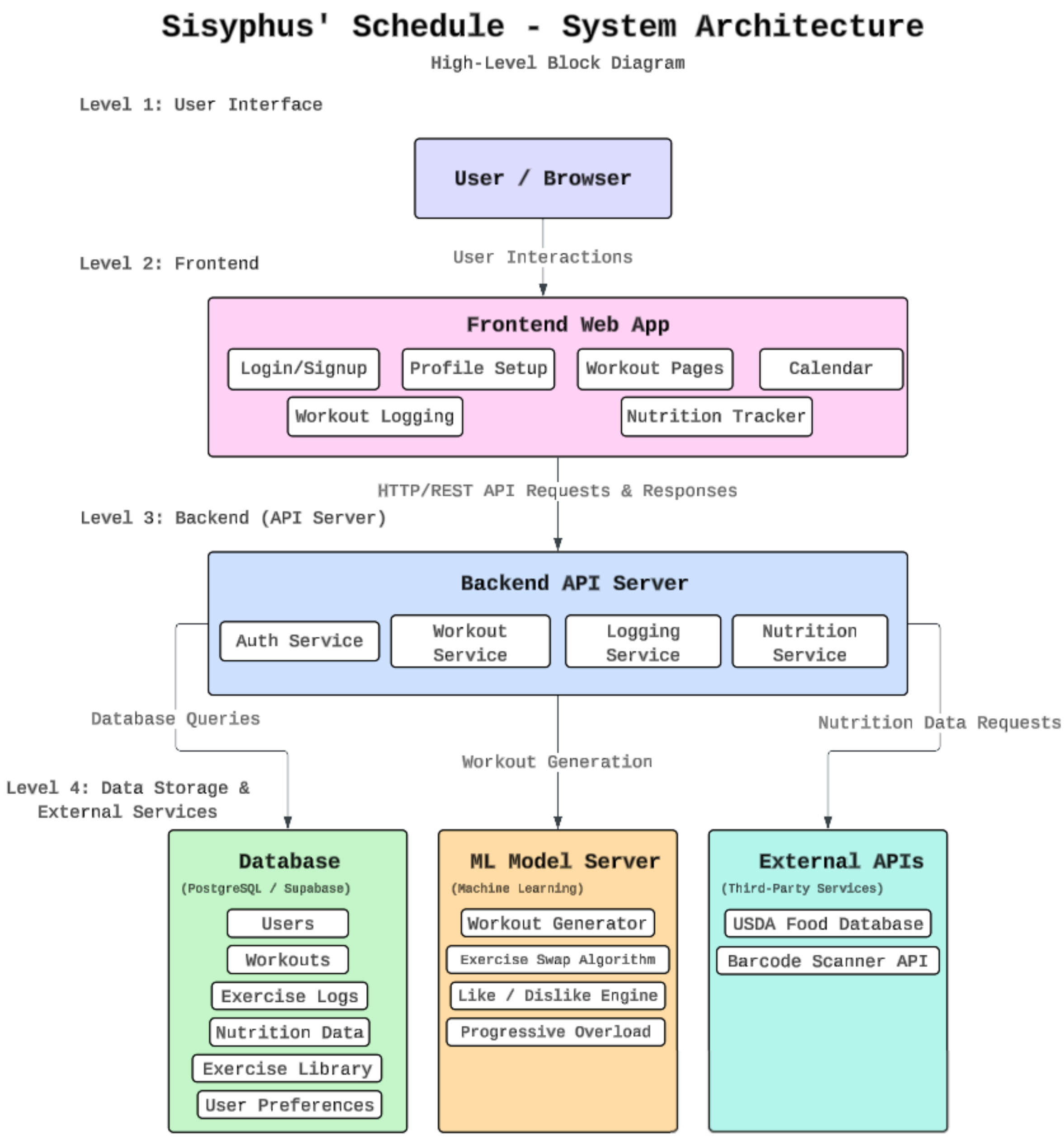


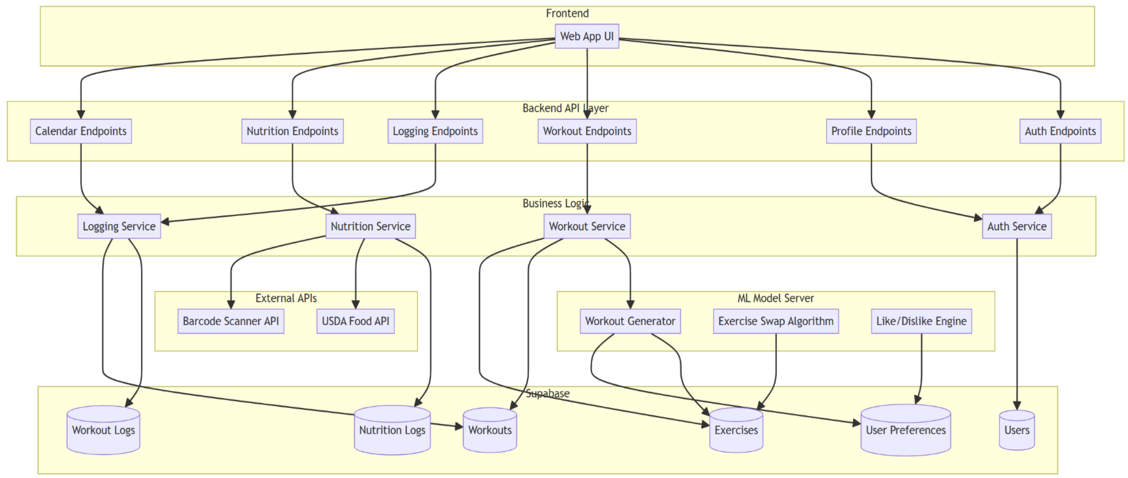
Option
No
Workouts
Data stored to show a library of workouts
Create a workout or browse workouts
Selected Page
Input email and password
Complete Questionnaire
Nutrition
Data stored from the USDA that holds information about stored and logged foods
Input foods or browse foods
Selected Page
Start
Sign-up
Decision
Selected page
Calendar/Schedule
Will hold the generated splits and workouts that our app had cated for the user
Select todays workout or future/previous workouts
Logout
No
Selected Page
Login
Decision
Start Workout
Yes
Yes
Start Page
Community
Share workouts and see what others are doing
View others workouts and select workouts you want to try
End
Selected Page
Settings
Login and profile data
Edit or update information
weight input
Yes
Decision: Like or Dislike Workout
Store Data from Workout and continue using lift for later sessions
No
Store weights, but delete workout from future sets
Finish Workout
No
Finish Workout
Complete Post-Workout Questionnaire
Yes



Flowchart shapes
Terminator
Start and end points
Process
An action or function
Decision
A question to be answered
Document
The input/output of a document
Data
Data available for input/output

6. A High-Level Block diagram showing the Major components and subcomponents of the project and how they fit together. (Need Front-end, ML, Back-end)





7. List and Detailed Description of the MAJOR Milestones for your project (think top 3-4 things) -- How will you know when you have successfully reached/completed each milestone?
    i. UI Designed - Frontend mostly complete. This means the majority of the interactable components such as buttons and pages are mostly complete without styling. Logic of page and feature navigation or modification outlined with flow charts.
    ii. Backend and Database Integrated - Users can sign up and log in against the real backend. Profile data, workouts, and logs are stored in the database. Frontend pages retrieve and display real data using API calls.
    iii. Having a completed exercise dataset and a sample workout model to feed into the model. To have all the algorithms fully functional to be able to take in data input and being able to generate data output to feed into other algorithms to generate workouts for users monthly. The Model should also be able to generate datasets for the disliked and liked exercises. When a user dislikes an exercise, it will look in the exercise dataset and pull another exercise that fits that same muscle group and action to replace it. The model should have a liked exercise dataset, and in those exercises on certain days, to keep the user engaged to keep using the app.
    iv. Beta testing: Giving the participants a before-and-after questionnaire to see what worked for them and what they did not like. With that questionnaire, we can make adjustments to the project and then release it again for new participants and old participants to see the changes. Give them another questionnaire to see any feedback and make adjustments to complete the project.

8. A List of major tasks that lead to each milestone (no details required); think in segments of ~2 week tasks for one person; set and assign the task to a lead for now (it might change later). This may be from Trello or another planning site, not required, but would be really helpful.
    - UI Designed - Tesla and Lily

1. Meet and organize what we want from our app as well as overall layout/design
2. Finish implementing mockups and flow diagrams to get proper functionality
3. Work with Base44 code to get a fully functioning application that contains our necessary pages and features.
    a. Change layout/design
    b. Make sure navigation and directory works

- User Feedback - Lily
    1. Get the user input on features they liked and disliked from the app
    2. Make changes based on that input
    3. Release it again to the users of the previous test and new users to get fresh eyes and experienced eyes.
    4. Based on the responses from this run, we will make those changes, and the project will be finalized
- Backend Core - Nolan and Thomas
    1. Design database schema for users, workouts, logs, nutrition.
    2. Implement auth, profile, workout, and logging endpoints.
    3. Connect frontend forms to backend APIs and start debugging.
- Machine Learning Model - Collin
    1. Get the completed exercise and workout dataset from our strength and conditioning advisor
    2. Develop it to create unique workouts based on questionnaire input
    3. Create an algorithm that will remove disliked exercises and replace them with something similar
    4. Prioritize the liked exercises to show up more in future workouts to keep the user engaged
    5. Have the model generate workouts on a monthly basis

9. A list of any additional/stretch goals that could be added to the project if time allows.
    - Community aspect for users to participate with friends:
        i. Add friends from their usernames
        ii. Worldwide and local leaderboards
        iii. Discovery page where you meet others that lift at your local gyms
            1. Optional disclosure of location.
        iv. Database of workouts from other users that you can query
    - Create a visual tutorial incorporated in the app to walk users through the different functions of the web app.
    - Automate progressive overload where it'll adjust sets, weight, and/or reps based on what the user is able to achieve.

- Add functionality to guide the user through creating long term training blocks composed of shorter term focused blocks. This could also include managing fatigue and frequency.

Include all in a single document that you would feel comfortable handing off to another team (of similar background/skill set) who would then go build out your project.

The document will be PDF named: final.pdf. Place the document in /planning directory and submit a link.